Online Examination Report

Date:	21.07.2014
Attempts:	1
Examinee:	ATIALUK, Joseph
Date of birth:	12/24/1987
Subject:	080-Principles of Flight
Result:	25 %
Time used:	00:00:13 (HH:MM:SS)

Attachments

1. List of result by question
2. List of result by topic
3. Detailed question result report

Attachment 1: List of result by question

CQB Number	Internal Number	Syllabus reference	Chapter	Points	Max. Points
1	214	81.	PRINCIPLES OF FLIGHT AEROPLANE	0,00	1,00
2	218	81.	PRINCIPLES OF FLIGHT AEROPLANE	0,00	1,00
3	217	81.	PRINCIPLES OF FLIGHT AEROPLANE	0,00	1,00
4	213	81.	PRINCIPLES OF FLIGHT AEROPLANE	0,00	1,00
5	147	81.	PRINCIPLES OF FLIGHT AEROPLANE	1,00	1,00
6	149	81.	PRINCIPLES OF FLIGHT AEROPLANE	0,00	1,00
7	211	81.	PRINCIPLES OF FLIGHT AEROPLANE	0,00	1,00
8	197	81.	PRINCIPLES OF FLIGHT AEROPLANE	1,00	1,00
9	233	81.	PRINCIPLES OF FLIGHT AEROPLANE	0,00	1,00
10	209	81.	PRINCIPLES OF FLIGHT AEROPLANE	0,00	1,00
11	228	81.	PRINCIPLES OF FLIGHT AEROPLANE	0,00	1,00
12	130	81.	PRINCIPLES OF FLIGHT AEROPLANE	1,00	1,00
13	193	81.	PRINCIPLES OF FLIGHT AEROPLANE	0,00	1,00
14	239	81.	PRINCIPLES OF FLIGHT AEROPLANE	1,00	1,00
15	222	81.	PRINCIPLES OF FLIGHT AEROPLANE	1,00	1,00
16	195	81.	PRINCIPLES OF FLIGHT AEROPLANE	0,00	1,00
17	85	81.1.1.1	Laws and definitions	0,00	1,00
18	82	81.1.1.1	Laws and definitions	0,00	1,00
19	17	81.1.1.5	Wing shape	0,00	1,00
20	30	81.1.2	Two-dimensional airflow around an aerofoil	0,00	1,00
21	27	81.1.2.2	Stagnation point	0,00	1,00
22	23	81.1.2.4	Centre of pressure and aerodynamic centre	0,00	1,00
23	84	81.1.3.1	The lift coefficient Cl	0,00	1,00
24	93	81.1.4.2	Induced drag	0,00	1,00
25	57	81.1.4.2	Induced drag	1,00	1,00
26	74	81.1.4.2	Induced drag	0,00	1,00
27	4	81.1.4.2	Induced drag	0,00	1,00
28	12	81.1.8.2	The stall speed	1,00	1,00

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CQB Number	Internal Number	Syllabus reference	Chapter	Points	Max. Points
29	52	81.2.2.1	Normal shock waves	0,00	1,00
30	104	81.4	STABILITY	0,00	1,00
31	86	81.4.1.1	Basics and definitions	1,00	1,00
32	14	81.4.1.1	Basics and definitions	0,00	1,00
33	79	81.4.3.7	Factors affecting the Cm - alpha graph	0,00	1,00
34	5	81.4.5.5	Factors affecting static lateral stability	0,00	1,00
35	32	81.5.8.3	Stabiliser trim	0,00	1,00
36	96	81.6.1.1	Flutter	0,00	1,00
37	110	81.7	PROPELLERS	1,00	1,00
38	109	81.7	PROPELLERS	0,00	1,00
39	76	81.7.1.1	Relevant propeller parameters	0,00	1,00
40	19	81.8.1.4	Straight steady glide	1,00	1,00
			Total:25%	**10,00**	**40,00**

Attachment 2: List of result by topic

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Date: 17.07.2014

Examinee: ATIALUK, Joseph

Location: Uganda Civil Aviation Authority

Your Result: 10,00 from 40,00 Points (25,00 %)

Topics	Ammount of Questions	Points achieved	Maximum Points	Percent
81. PRINCIPLES OF FLIGHT AEROPLANE	40	10,00	40,00	25,00
81.1. SUBSONIC AERODYNAMICS	12	2,00	12,00	16,67
81.2. HIGH SPEED AERODYNAMICS	1	0,00	1,00	0,00
81.4. STABILITY	5	1,00	5,00	20,00
81.5. CONTROL	1	0,00	1,00	0,00
81.6. LIMITATIONS	1	0,00	1,00	0,00
81.7. PROPELLERS	3	1,00	3,00	33,33
81.8. FLIGHT MECHANICS	1	1,00	1,00	100,00
Total	**40**	**10,00**	**40,00**	**25,00**

1

What is the suggested speed to fly when passing through lift with no intention to work the lift?

- ☐ **Best lift/drag speed.**
- ☐ **Minimum sink speed.**
- ☒ **Best glide speed.**

ID:	214(LTS:226395, LTM:214)	Max. Points	1,00
Syllabus reference:	81. PRINCIPLES OF FLIGHT AEROPLANE	Points	0,00

2

The practice of allowing the ground crew to lift a balloon into the air is

- ☐ **a safe way to reduce stress on the envelope.**

- ☒ **considered to be good practice, particularly when obstacles must be cleared shortly after lift-off.**

- ☐ **unsafe because it can lead to a sudden landing at an inopportune site just after lift-off.**

ID:	218(LTS:226399, LTM:218)	Max. Points	1,00
Syllabus reference:	81. PRINCIPLES OF FLIGHT AEROPLANE	Points	0,00

3

What is the relationship of false lift to the wind? False lift

- [x] increases if the vertical velocity of a balloon increases.
- [ ] decreases as the wind accelerates a balloon to the same speed as the wind.
- [ ] exists only if the surface winds are calm.

ID:	217(LTS:226398, LTM:217)	Max. Points	1,00
Syllabus reference:	81. PRINCIPLES OF FLIGHT AEROPLANE	Points	0,00

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4

Which is true relating to the direction in which turns should be made during slope soaring?

☒ **All reversing turns should be made to the left.**

☐ **All reversing turns should be made into the wind away from the slope.**

☐ **All turns should be made downwind toward the slope.**

ID:	213(LTS:226394, LTM:213)	Max. Points	1,00
Syllabus reference:	81. PRINCIPLES OF FLIGHT AEROPLANE	Points	0,00

5

When a slight upward or negative flap deflection is used, the result is

☐ **increased drag.**

☐ **decreased lift.**

☒ **decreased drag.**

ID: 147(LTS:226359, LTM:147)
Syllabus reference: 81. PRINCIPLES OF FLIGHT AEROPLANE

Max. Points 1,00
Points 1,00

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6

At what bank angle will the resultant of gravity and centrifugal force equal twice a glider's weight?

- ☐ **30 degrees.**
- ☐ **60 degree.**
- ☒ **45 degrees.**

ID:	149(LTS:226361, LTM:149)	Max. Points	1,00
Syllabus reference:	81. PRINCIPLES OF FLIGHT AEROPLANE	Points	0,00

7

If swirling dust, leaves, or debris indicate a strong thermal on the final approach to a landing, it is recommended that the glider pilot

- ☐ close the spoilers and increase the airspeed.
- ☒ open the spoilers and reduce the airspeed.
- ☐ open the spoilers and maintain a constant airspeed.

ID:	211(LTS:226392, LTM:211)	Max. Points	1,00
Syllabus reference:	81. PRINCIPLES OF FLIGHT AEROPLANE	Points	0,00

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8

If the gyroplane's CG is below the propeller thrust line, which direction will the application of power cause the nose to move?

- [x] **The nose will pitch down.**
- [ ] **The nose will not move.**
- [ ] **The nose will pitch up.**

ID:	197(LTS:226378, LTM:197)	Max. Points	1,00
Syllabus reference:	81. PRINCIPLES OF FLIGHT AEROPLANE	Points	1,00

9

After being handed off to the final approach controller during a 'no-gyro' surveillance or precision approach, the pilot should make all turns

- ☒ **based upon the groundspeed of the aircraft.**
- ☐ **standard rate.**
- ☐ **one-half standard rate.**

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10

Unless adequate speed control is maintained during the turn to base and the final approach for a landing into the wind, which would most likely occur if a steep wind gradient existed?

- ☐ The wingtip on the outside of the turn would stall before the wingtip on the inside of the turn.
- ☒ The airspeed on final approach would increase, causing the glider to overshoot the desired landing spot.
- ☐ The desired landing spot would be undershot or the glider would stall.

ID:	209(LTS:226390, LTM:209)	Max. Points	1,00
Syllabus reference:	81. PRINCIPLES OF FLIGHT AEROPLANE	Points	0,00

11

The keel pocket's purpose is to

- ☐ act as a roll stabilizer, keeping the wing from wandering left and right.
- ☐ act as a yaw stabilizer, keeping the wing from wandering left and right.
- ☒ act as a longitudinal stabilizer, keeping the wing from wandering left and right.

ID:	228(LTS:226408, LTM:228)	Max. Points	1,00
Syllabus reference:	81. PRINCIPLES OF FLIGHT AEROPLANE	Points	0,00

12

What type of stability does the horizontal stabilizer provide during flight?

- [x] **Longitudinal.**
- [ ] **Airspeed.**
- [ ] **Lateral.**

ID:	130(LTS:226342, LTM:130)	Max. Points	1,00
Syllabus reference:	81. PRINCIPLES OF FLIGHT AEROPLANE	Points	1,00

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The stalling speed of an aircraft will be highest when the aircraft is loaded with a

☒ **high gross weight and aft CG.**

☐ **high gross weight and forward CG.**

☐ **low gross weight and forward CG.**

ID:	193(LTS:226374, LTM:193)	Max. Points	1,00
Syllabus reference:	81. PRINCIPLES OF FLIGHT AEROPLANE	Points	0,00

14

(Refer to Figure 145.) What is the correct sequence for recovery from the unusual attitude indicated?

See Attachments:
1. instrument_145.jpg figure 245 (Attachment No.1)

- ☐ **Reduce power, increase back elevator pressure, and level the wings.**
- ☐ **Level the wings, raise the nose of the aircraft to level flight attitude, and obtain desired airspeed.**
- ☒ **Reduce power, level the wings, bring pitch attitude to level flight.**

ID:	239(LTS:226419, LTM:239)	Max. Points	1,00
Syllabus reference:	81. PRINCIPLES OF FLIGHT AEROPLANE	Points	1,00

15

Which stall must be performed during a flight instructor - airplane practical test?

☐ **Accelerated.**

☐ **Imminent.**

☒ **Power-on or power-off.**

ID:	222(LTS:226403, LTM:222)	Max. Points	1,00
Syllabus reference:	81. PRINCIPLES OF FLIGHT AEROPLANE	Points	1,00

16

When an aircraft's forward CG limit is exceeded, it will affect the flight characteristics of the aircraft by producing

- [x] **improved performance since it reduces the induced drag.**
- [ ] **very light elevator control forces which make it easy to inadvertently overstress the aircraft.**
- [ ] **higher stalling speeds and more longitudinal stability.**

ID:	195(LTS:226376, LTM:195)	Max. Points	1,00
Syllabus reference:	81. PRINCIPLES OF FLIGHT AEROPLANE	Points	0,00

17

Which formula or equation describes the relationship between force (F), acceleration (a) and mass (m)?

- ☐ a = F * m
- ☐ m = F * a
- ☐ F = m * a
- ☒ F = m / a

ID:	85(LTS:226297, LTM:85)	Max. Points	1,00
Syllabus reference:	81.1.1.1 Laws and definitions	Points	0,00

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18

The SI units of air density (I) and force (II) are..... &... respectively:

- ☐ (I) kg / m², (II) kg.
- ☐ (I) N / kg, (II) kg.
- ☒ (I) N / m³, (II) N.
- ☐ (I) kg / m³, (II) N.

ID:	82(LTS:226294, LTM:82)	Max. Points	1,00
Syllabus reference:	81.1.1.1 Laws and definitions	Points	0,00

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19

Taper ratio of a wing is the ratio between:

- ☐ **root chord and tip chord.**
- ☒ **wing span squared and wing area.**
- ☐ **mean geometric chord and wing span.**
- ☐ **tip chord and root chord.**

ID:	17(LTS:226229, LTM:17)	Max. Points	1,00
Syllabus reference:	81.1.1.5 Wing shape	Points	0,00

20

(Please view annex 081-0005 issue date July 2004)
How are the speeds at point 1 and point 2 related in the figure to the relative wind/airflow V?

See Attachments:

1. 081-0005.gif Annex (Attachment No.2)

- ☐ **V1 = 0 and V2 = V.**
- ☐ **V1 < V2 and V2 < V.**
- ☐ **V1 = 0 and V2 > V.**
- ☒ **V1 > V2 and V2 < V.**

ID: 30(LTS:226242, LTM:30)
Syllabus reference: 81.1.2 Two-dimensional airflow around an aerofoil

Max. Points 1,00
Points 0,00

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21

The stagnation point is the point:

☐ where the velocity of the relative airflow is reduced to zero.

☒ of the intersection of the thrust vector and the chord line.

☐ relative to which the sum of all moments is independent of angle of attack.

☐ of the intersection of the total aerodynamic force and the chord line.

ID:	27(LTS:226238, LTM:27)	Max. Points	1,00
Syllabus reference:	81.1.2.2 Stagnation point	Points	0,00

22

Assuming no flow separation and no compressibility effects the location of the centre of pressure of a symmetrical aerofoil section:

- ☐ **moves forward when the angle of attack decreases.**
- ☒ **moves backward when the angle of attack decreases.**
- ☐ **is at approximately 50% chord irrespective of angle of attack.**
- ☐ **is at approximately 25% chord irrespective of angle of attack.**

ID:	23(LTS:226234, LTM:23)	Max. Points	1,00
Syllabus reference:	81.1.2.4 Centre of pressure and aerodynamic centre	Points	0,00

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23

The lift formula can be written as:
(rho = density)

- ☒ L = W.
- ☐ L = n * W.
- ☐ L = CL * ½rho * V² * S.
- ☐ L = CL * 2rho * V² * S.

ID: 84(LTS:226296, LTM:84)
Syllabus reference: 81.1.3.1 The lift coefficient Cl

Max. Points 1,00
Points 0,00

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24

Which of these statements about induced drag are correct or incorrect?

I. An elliptical spanwise lift distribution generates more induced drag than a rectangular lift distribution.
II. Induced drag decreases with decreasing aspect ratio.

- ☐ I is correct, II is incorrect.
- ☐ I is incorrect, II is incorrect.
- ☐ I is incorrect, II is correct.
- ☒ I is correct, II is correct.

ID:	93(LTS:226305, LTM:93)	Max. Points	1,00
Syllabus reference:	81.1.4.2 Induced drag	Points	0,00

25

The induced drag:

- ☐ increases as the aspect ratio increases.
- ☐ increases as the magnitude of the tip vortices decreases.
- ☐ has no relation to the lift coefficient.
- ☒ increases as the lift coefficient increases.

ID:	57(LTS:226269, LTM:57)	Max. Points	1,00
Syllabus reference:	81.1.4.2 Induced drag	Points	1,00

26

An aeroplane transitions from steady straight and level flight into a horizontal co-ordinated turn with a load factor of 2, the speed remains constant and the:

- ☐ **angle of attack increases by a factor of 1/4.**
- ☐ **total drag increases by a factor of 4.**
- ☐ **induced drag increases by a factor of 4.**
- ☒ **lift increases by a factor of 4.**

ID: 74(LTS:226286, LTM:74)
Syllabus reference: 81.1.4.2 Induced drag

Max. Points 1,00
Points 0,00

27

Which statement concerning the local flow pattern around a wing is correct?

- ☒ **Slat extension, at a constant angle of attack and normal extension speeds, will increase the lift coefficient,**
- ☐ **Vortex generators on the wing partially block the spanwise flow over the wing**
- ☐ **By fitting winglets to the wing tip,reduces induced drag.**
- ☐ **Sweepback reduces drag since, because the span increases.**

ID:	4(LTS:226216, LTM:4)	Max. Points	1,00
Syllabus reference:	81.1.4.2 Induced drag	Points	0,00

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The stall speed:

- ☐ decreases with an increased weight.
- ☐ increases with the length of the wingspan.
- ☐ does not depend on weight.
- ☒ increases with an increased weight.

ID:	12(LTS:226224, LTM:12)	Max. Points	1,00
Syllabus reference:	81.1.8.2 The stall speed	Points	1,00

29

Which of the following statements is correct?
Air passes a normal shock wave as

- ☒ **The static pressure decreases.**
- ☐ **The velocity increases.**
- ☐ **The static temperature decreases.**
- ☐ **The static temperature increases.**

ID:	52(LTS:226264, LTM:52)	Max. Points	1,00
Syllabus reference:	81.2.2.1 Normal shock waves	Points	0,00

30

If an aircraft has negative dynamic and positive static stability, this will result in

- ☐ undamped oscillations.
- ☒ convergent oscillations.
- ☐ divergent oscillations.

ID:	104(LTS:226316, LTM:104)	Max. Points	1,00
Syllabus reference:	81.4 STABILITY	Points	0,00

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31

Positive static stability of an aeroplane means that following a disturbance from the equilibrium condition:

- [x] **the initial tendency is to return towards its equilibrium condition.**
- [ ] **the tendency is to move with an oscillatory motion of decreasing amplitude.**
- [ ] **the tendency is to move with an oscillatory motion of increasing amplitude.**
- [ ] **the initial tendency is to diverge further from its equilibrium condition.**

ID:	86(LTS:226298, LTM:86)	Max. Points	1,00
Syllabus reference:	81.4.1.1 Basics and definitions	Points	1,00

32

A statically unstable aeroplane:

- ☐ is always dynamically stable.
- ☐ reduces its instability as the CG moves aft.
- ☐ can never be dynamically stable.
- ☒ will become stable at high speed.

ID:	14(LTS:226226, LTM:14)	Max. Points	1,00
Syllabus reference:	81.4.1.1 Basics and definitions	Points	0,00

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See Attachments:

1. 081-0027.gif Annex (Attachment No.3)

33

(Please view annex 081-0027 issue date July 2006)
Which of these statements about the pitching moment coefficient versus angle of attack lines in the annex is correct?

- ☐ **The CG position is further forward at line 1 when compared with line 4.**
- ☐ **Static longitudinal stability is greater at line 3 when compared with line 4 at low and moderate angles of attack.**
- ☐ **The CG position is further aft at line 1 when compared with line 4.**
- ☒ **In its curved part at high angles of attack line 2 illustrates a decreasing static longitudinal stability.**

ID:	79(LTS:226291, LTM:79)	Max. Points	1,00
Syllabus reference:	81.4.3.7 Factors affecting the Cm - alpha graph	Points	0,00

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34

Sweepback of a wing positively influences:

1. static longitudinal stability.
2. static lateral stability.
3. dynamic longitudinal stability.

The combination that regroups all of the correct statements is:

☒ **1, 2, 3.**

☐ **3.**

☐ **2.**

☐ **1.**

ID: 5(LTS:226217, LTM:5)
Syllabus reference: 81.4.5.5 Factors affecting static lateral stability

Max. Points 1,00
Points 0,00

35

When comparing a stabiliser trim system with an elevator trim system, which of these statements is correct?

- ☐ a stabiliser trim is more sensitive to flutter.
- ☒ an elevator trim is able to compensate larger changes in pitching moments.
- ☐ an elevator trim is more suitable for aeroplanes with a large CG range.
- ☐ a stabiliser trim is able to compensate larger changes in pitching moments.

ID:	32(LTS:226244, LTM:32)	Max. Points	1,00
Syllabus reference:	81.5.8.3 Stabiliser trim	Points	0,00

36

Wing flutter may be caused by a:

- ☐ combination of bending and torsion of the wing structure.
- ☐ combination roll control reversal and low speed stall.
- ☒ aerodynamic wing stall at high speed.
- ☐ combination of fuselage bending and wing torsion.

ID:	96(LTS:226308, LTM:96)	Max. Points	1,00
Syllabus reference:	81.6.1.1 Flutter	Points	0,00

37

The reason for variations in geometric pitch (twisting) along a propeller blade is that it

- [x] **permits a relatively constant angle of attack along its length when in cruising flight.**
- [ ] **prevents the portion of the blade near the hub to stall during cruising flight.**
- [ ] **permits a relatively constant angle of incidence along its length when in cruising flight.**

ID:	110(LTS:226322, LTM:110)	Max. Points	1,00
Syllabus reference:	81.7 PROPELLERS	Points	1,00

38

A propeller rotating clockwise, as seen from the rear, creates a spiraling slipstream that tends to rotate the aircraft to the

☒ right around the vertical axis, and to the left around the longitudinal axis.

☐ left around the vertical axis, and to the left around the longitudinal axis.

☐ left around the vertical axis, and to the right around the longitudinal axis.

ID:	109(LTS:226321, LTM:109)	Max. Points	1,00
Syllabus reference:	81.7 PROPELLERS	Points	0,00

39

The difference between a propeller's blade angle and its angle of attack is called:

- ☐ **the helix angle.**
- ☐ **propeller slip.**
- ☒ **the propeller angle.**
- ☐ **the effective pitch.**

ID:	76(LTS:226288, LTM:76)	Max. Points	1,00
Syllabus reference:	81.7.1.1 Relevant propeller parameters	Points	0,00

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40

The descent angle of a given aeroplane in a steady wings level glide has a fixed value for a certain combination of:

- ☐ **mass and altitude.**
- ☐ **configuration and mass.**
- ☒ **configuration and angle of attack.**
- ☐ **altitude and configuration.**

ID:	19(LTS:226231, LTM:19)	Max. Points	1,00
Syllabus reference:	81.8.1.4 Straight steady glide	Points	1,00

Attachment No. 1

Questions 14
figure 245
instrument_145.jpg

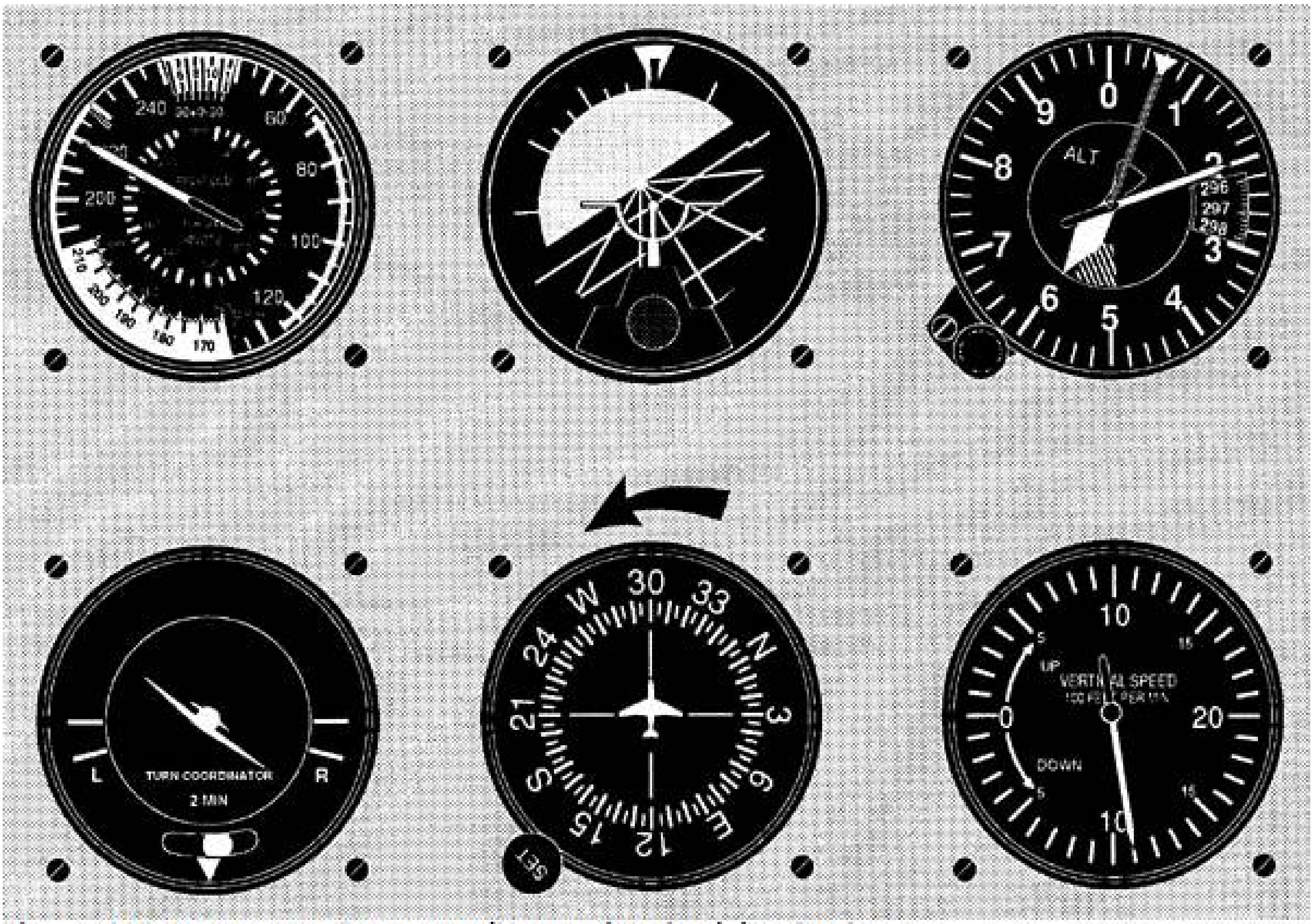


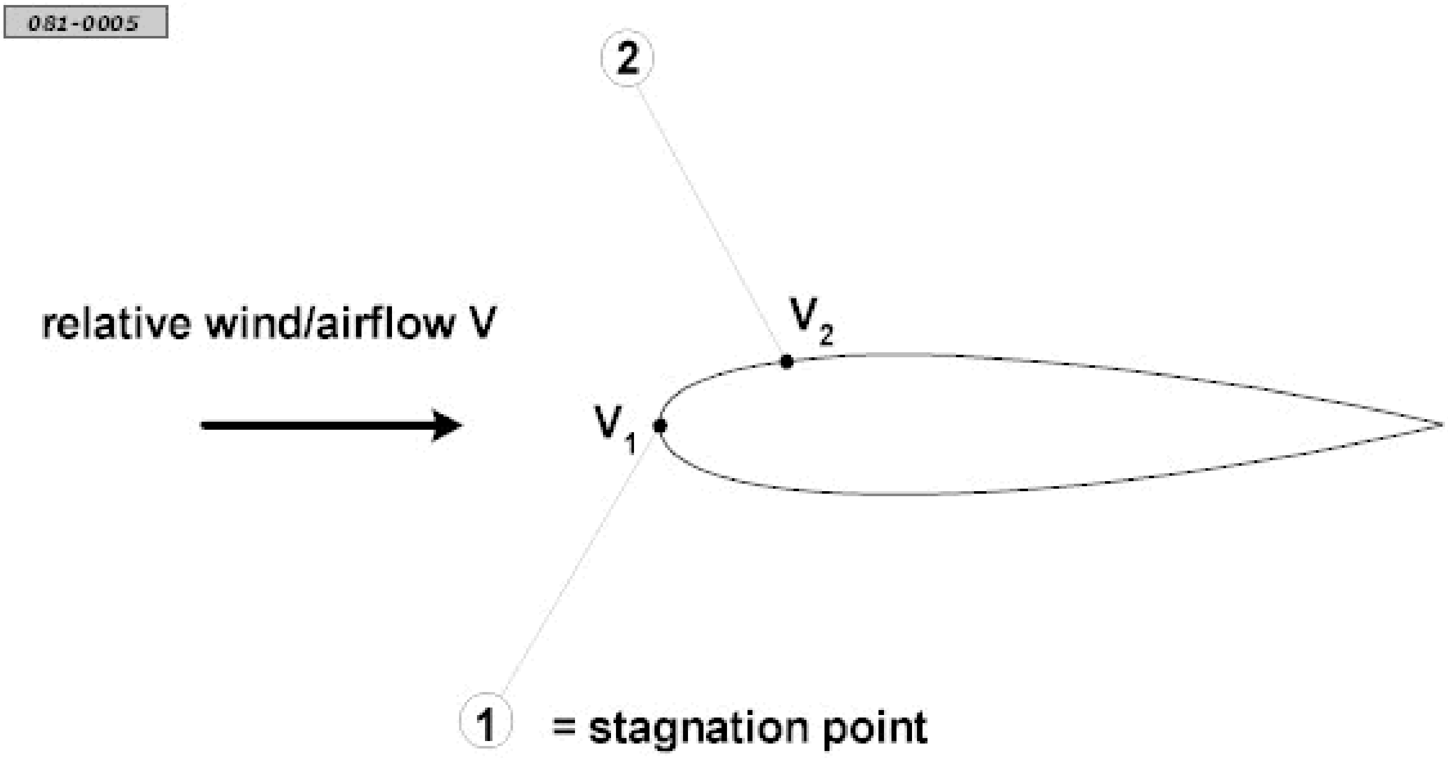
Figure 145. Instrument Sequence (Unusual Attitude) © ASA

Attachment No. 2

Questions 20

Annex

081-0005.gif



Attachment No. 3

Questions 33

Annex

081-0027.gif